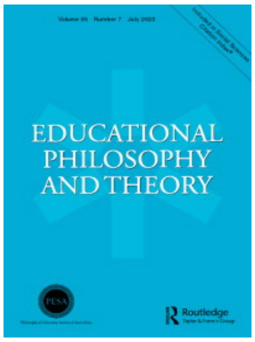

Educational Philosophy and Theory

ISSN: 0013-1857 (Print) 1469-5812 (Online) Journal homepage: www.tandfonline.com/journals/rept20

Digital trade, digital economy and the digital economy partnership agreement (DEPA)

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EDUCATIONAL PHILOSOPHY AND THEORY
2023, VOL. 55, NO. 7, 747–755
https://doi.org/10.1080/00131857.2022.2041413

EDITORIAL

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Introduction

From the beginning of the new century developmentalist economists indicated the importance of the coming digital economy based on information, computing and communication, including reference to the profound effects of the internet and the web on the organization of firms and the linkages and connections between the local and the global (Brynjolfsson & Kahin, 2002). Twenty years later analysts are talking freely about the internet or web economy and trying to measure interconnectivities in terms of crowdsourcing, smart energy design, and the reconstruction of the global value chain to explore 'intelligent capitalism'. The emphasis has fallen on financialization and the growth of next generation strategic digital technologies of deep learning, artificial intelligence, quantum computing, 5 G/6G, and bioinformatics that operate at the nanolevel to promote technological convergence and lead the next phase of globalization (Peters, 2020). Business innovation, trade facilitation, and development are seen to depend on leveraging consumer psychology and of promoting scale economies within what is referred to as 'platform capitalism'. Increasingly social media practices collapse the public and the personal, and hinge on the costs of storing, sharing, and analysing data which has decreased with increased activity in the Cloud as the internet continues to evolve. The digital economy is seen as providing a new architecture for a transformation of the labour market based on job growth, competencies and skills in the digital economy. There is a new respect for social networks and service ecosystems in the age of networked intelligence and consumption-based forecasting. Of fundamental importance is the digital transformation of finance, large investment banks, big asset managers, and hedge fund management but also the new group of sovereign investors that have emerged as influential capital markets players and investment firms within a $30 trillion global market.

At the same time, under Covid pandemic conditions the digital economy has become a potent driving force in the world economy particularly for service industries that utilise symbolic goods and services that can be reduced to highly transportable exchange and circulation such as services of trade facilitation, banking and insurance, software, entertainment, scientific research and publishing and international education. As Yue et al. (2020) explain the concept of 'digital economy' was first used in the 1990s developed a global focus at the 'G20 Digital Economy Development and Cooperation Initiative' at the G20 Summit in Hangzhou, the digital economy has 'become a new focus of China's economic development and is becoming a policy focus: 'China's economic development is facing a "three-phase superposition" period of speed shift, structural adjustment and power conversion, thus vigorous digital economy has become a new driving force for China's economy and an important force leading the national innovation strategy' (p. 1). The G20 conference was encouraged to 'collectively leverage digital opportunities, cope with challenges, and promote the digital economy to drive inclusive economic growth and development'.[1] The final statement provides a useful description of the digital economy that

> refers to a broad range of economic activities that include using *digitized information and knowledge as the key factor of production*, modern information networks as an important activity space, and the effective use of information and communication technology (ICT) as an important driver of productivity growth and economic structural optimization. Internet, cloud computing, big data, Internet of Things (IoT), fintech and other new digital technologies are used to collect, store, analyze, and share information digitally and transform social interactions. Digitized, networked and intelligent ICTs enable modern economic activities to be more flexible, agile and smart (p. 1).

This final statement was drafted in 2016 several years before the Covid pandemic accelerated the use of digital technology, digital trade and resilient ICTs. The statement also recognized the guiding 'common principles' as: innovation, partnership, synergy, flexibility, inclusion; open and enable business environment, and 'Flow of Information for Economic Growth, Trust and Security' (p. 2). At this meeting six years ago the G20 indicated: 'Recognizing that the digital shifts underway are reshaping economies and societies today and will continue to do so in the future, the G20 agrees to cooperate and continue to work closely on these matters' (p. 8).

Foremost in contemporary rethinking of the digital economy is the prospect and reality of digital trade where the internet serves as a trade platform that raises new issues concerning the free flow of data across borders, security and privacy, the protection of source code and digital localization. The Trans-Pacific Partnership (TPP), for instance, addressed digital trade in terms of a set of 'obligations designed to promote the digital economy through a free and open Internet and commerce without borders', including an emphasis on the elimination of digital customs duties and forced technology transfers, fair and non-discriminatory network competition, cooperation on cybersecurity, promotion of innovation encryption, authentication, and global interoperability, strong patent and copyright protections and measures to prevent trade secret theft (Table 1).[2]

As Jim Foster (2016) indicates TPP is the first multilateral agreement 'to specifically take up economic and trade issues related to the Internet and includes ground-breaking provisions with respect to online privacy, cybersecurity, promotion of cross-border data flows and competition policy'. As he goes on to explain 'The new disciplines within the TPP set out rules that cut across all sectors of the economy premised on the role of "data" as a new mode of production'

Table 1. TPP principles of digital trade.

1. Promoting a free & open internet
2. Prohibiting digital customs duties
3. Securing basic non-discrimination principles
4. Enabling cross-border data flows
5. Preventing localization barriers
6. Barring forced technology transfers
7. Protecting critical source code
8. Ensuring technology choice
9. Advancing innovative authentication methods
10. Delivering enforceable consumer protections
11. Safeguarding network competition
12. Fostering innovative encryption products
13. Building an adaptable framework for digital trade
14. Promoting cooperation on cybersecurity
15. Preserving market-driven standardization & global interoperability
16. Eliminating tariffs on all manufactured products
17. Securing robust market access commitments on investment & crossborder services, including those delivered digitally
18. Ensuring faster, more transparent customs procedures
19. Promoting transparency & stakeholder participation in the development of regulations & standards
20. Ensuring fair competition with state-owned enterprises
21. Promoting strong & balanced copyright protections & enforcement
22. Advancing modern patent protection
23. Combatting trade secret theft
24. Recognizing conformity assessment procedures

Source: Compiled from https://ustr.gov/sites/default/files/Digital-2-Dozen-Final.pdf.

(p. 1). New rules have to be established for privacy, intellectual property, cyber security, government procurement and localization, with TPP support for greater utilization of ICT in the economies of the region with help to overcome challenges of the digital divide and diversity. Foster indicates that TPP is a 'game-changer' and suggests that TPP protocols are likely to be performance-based rather than prescriptive.

The digital economic and digital trade has rapidly expanded and all evidence points to the spectacular growth of the internet economy in an age of Covid as e-commerce has bloomed. A Bloomberg-based article in *The Bangkok Post* carries the headline 'SE Asia digital economy to reach $363bn by 2025' eclipsing the previous forecast of $300 billion:

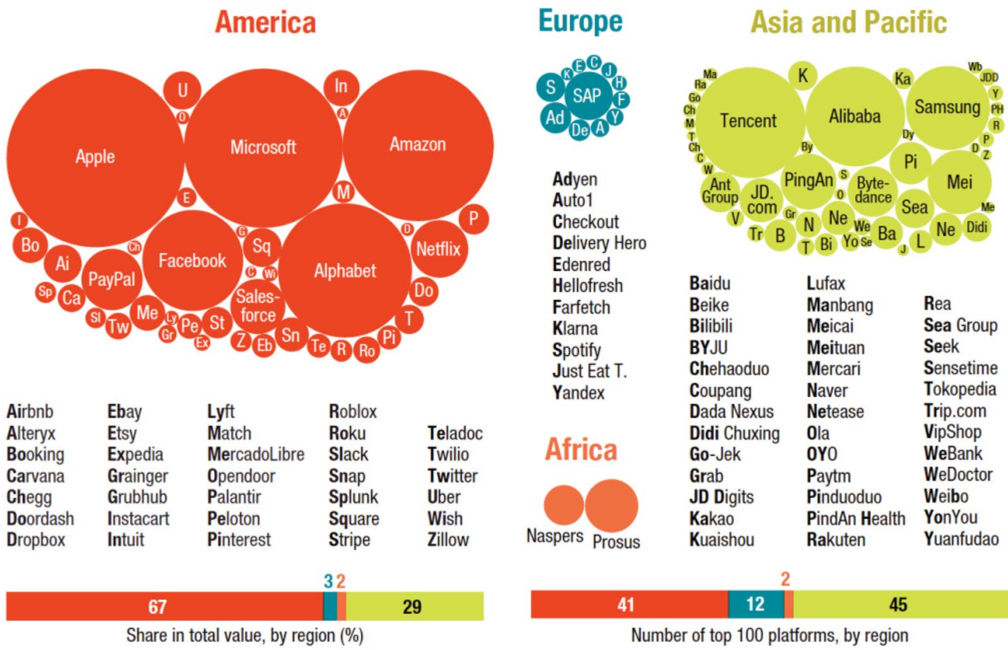
> E-commerce, travel, media, transport and food are driving the region's digital growth, with online spending rising 49% in 2021 to $174 billion, the companies said in their latest annual report. The region added 60 million new digital consumers since the start of the pandemic, led by Thailand and the Philippines.
>
> Southeast Asia, home to Alibaba Group Holding Ltd's Lazada and Tencent Holdings Ltd-backed Sea Ltd, will see a 62% increase in e-commerce gross merchandise value (GMV) this year as home-bound consumers pick up groceries and essentials from the likes of Lazada's RedMart and Sea's Shopee. Online shopping is now forecast to hit $234 billion in 2025 versus a previous $172 billion estimate, making up 64% of the region's total estimated digital GMV of $363 billion, the research shows.[2]

Vice president at Google Southeast Asia, Stephanie Davis is reported as saying 'Digital consumerism has really taken hold since the pandemic. We're seeing a digital decade ahead of us in Southeast Asia and a possibility of $1 trillion economy in the digital space by 2030.' The article also refers to Rohit Sipahimalani, chief investment strategist and head of Southeast Asia at Temasek, who indicates that 'Indonesia is the region's largest digital economy where online spending will probably double to $146 billion by 2025. Venture capital deals in the country in the first half of 2021 surpassed the full-year totals of each of the past four years' and 'Vietnam is expected to grow at the fastest rate among the six countries tracked by the study, nearly tripling in online GMV over the next four years'. Clearly, the digital economy of Southeast Asia is becoming a region where global scaling is relatively easily achieved. What is typical of Southeast Asia also has application to China, India and Asia as a whole.

UNCTAD's (UNCTAD 2021) timely *The Digital Economy Report 2021* examines 'the development and policy implications of cross-border flows of digital data' that are 'core to all fast-evolving digital technologies, such as data analytics, artificial intelligence (AI), blockchain, Internet of Things (IoT), cloud computing and other Internet-based services.' The data-driven global economy with the huge expansion of cross-border data flows impacts on digital trade and the Sustainable Development Goals yet it seems that it is a new area of policy and governments and agencies are unclear what approach they should follow. The stakes are high and the consequences will be determining for economic progress. It is clear that China and the US are the prime movers accounting for over 90% of all AI start-ups and most AI researchers, and the largest digital platforms – the Big Five in US (Apple, Microsoft, Amazon, Alphabet, Facebook) and the Big Three in China (Tencent- We-Chat, Alibaba, Baidu) – are busy investing in the data value chains including submarine transmission, storage in the cloud, AI analysis and processing from data collection to data intelligence.

As the report makes clear the 'data value chain' is a key concept for assessing value in the transformation of data, yet as it makes clear 'Despite the importance of data in the evolving digital economy, there is no universally agreed understanding of the concept of data, which may lead to confusion and increase complexity in analyses and policy debates.' The critical understanding is that '*Data are a special resource, with specific characteristics that make them different from goods and services. They are intangible and non-rival, which means that many people can use the same data simultaneously, or over time, without depleting them'* (my emphases).

Geographical distribution of the top 100 global digital platforms, by market capitalization 2021 (UNCTAD, 2021:22)



Source: Holger Schmidt, available at www.netzoekonom.de/vortraege/#tab-id-1 (data as of May 2021).
Note: As a reference, the market capitalization of Apple is $2.22 trillion, while for Mercado Libre it is $88.7 billion, $80.2 billion for Baidu and $59.7 billion for Spotify.

The analysis of data and data flows requires different perspectives and taxonomies:

> **First,** there has always been data and information associated with commercial transactions – such as billing data, banking data, names and delivery addresses – which are mainly volunteered and rarely create policy related issues, as long as new digital economy players work by the same rules as the conventional economy. **Second,** raw data gathered from individual activities, products, events and behaviours have no value in themselves, but can generate value once aggregated, processed and monetized, or used for social purposes. **Third**, the processing of raw data into digital intelligence – in the form of statistics, databases, insights, information, etc. – results in "data products", which may be considered as services in trade statistics when sold across borders. (UNCTAD 2021:99)

As the report comments rather than focusing on ownership, rights to access, control and use are more important. I regard shared data flows as part of the concept of knowledge socialism (Peters et al., 2020) that contrasts with an approach from intellectual property where data is controlled by the private sector by comparison with China that emphasizes government control and between these two alternatives the EU focuses on individual rights. The report notes that 'There is a race for leadership in technological developments, as the leader may gain an economic as well as a strategic advantage, by controlling the data and related technologies, particularly with regard to AI' which may well lead to the fragmentation of the digital realm and the development of dual or multiple internet ecologies.

The report in addition provides on ownership, access, control and rights over data with implications for cross-border implications for cross-border data flows; power imbalances and inequalities arising from these cross-border flows especially for developing countries; as well as major approaches to the digital economy and cross-border data flows and how they might promote market and innovation, public security and digital development. There is also due concern for cross-border data-flows trade agreements and the need for global data governance (Elsig et al. 2019).

Datafication and the data-driven economy

Data circulation, storage and analysis are increasingly one of the central features of contemporary capitalism. With its roots in economic capital data collection is

> driven by the perpetual cycle of capital accumulation, which in turn drives capital to construct and rely upon a universe in which everything is made of data. The imperative to capture all data, from all sources, by any means possible influences many key decisions about business models, political governance, and technological development (Sadowski, 2019).

The future of digital economy and digital trade requires a better understanding of international digital capitalism as a form of political economy. Various terms have emerged to capture this datafication of value chains, and ownership of metadigital systems that now dominate platform or cyber capitalism, and the structure of the internet and WWW. Data capitalism emerged in the 1990s to take a central role in the information and knowledge economies, understood as the shift from an e-commerce model to an advertising model premised on the sale of audiences, similar to the shift from print capitalism in the 19th century to media audience in the 2oth century (West, 2019).

Nick and Mejias (2019) talk of 'data colonialism' and how big data is appropriating, capturing and processing social data and converting everyday life into a continuous data stream, creating in its wake a new social order based on continuous tracking 'normalizing the exploitation of human beings through data, just as historic colonialism appropriated territory and resources and ruled subjects for profit' and paving the way for 'the capitalization of life without limit' (p. 336).

In strict contrast, the OECD (2021) provides a 'Digital Trade Inventory' to help 'countries navigate the complex and evolving digital trade landscape' and the new issues not covered by the regulatory environment covered by the WTO 'including e-signatures and e-payments, information flows, privacy, consumer protection, cybersecurity and market access.' As the OECD (2021) points out, adherence to international instruments varies widely across a number of jurisdictions: 'The Inventory identifies 52 instruments that have been developed across 24 different fora.'[3] The OECD indicates the fundamental significance of the flow of data as both a means of production and a tradeable asset that 'also underpins physical trade less directly by enabling implementation of trade facilitation. Data is also at the core of new and rapidly growing service supply models such as cloud computing, the Internet of Things (IoT), and additive manufacturing.'[4]

While there is no accepted definition of 'digital trade' it is clear that digitalization changes trade not only increasing 'the scale, scope and speed of trade' but expands global markets and enables firms to use digital technologies to develop new markets, facilitate electronic payments, utilize cloud services, and encourage greater collaboration and crowdfunding mechanisms.[5] Digital trade is advancing rapidly. It is becoming a means of increasing the size of global markets both for physical goods and services and offers the potential to strengthen the green supply chain and sustainability design. New blockchain technologies have the potential to disintermediate banking key services including clearance and settlement systems, fundraising, securities, loans and credit, transforming the financial services ecosystem. I have already mentioned capital formation through crowdfunding. In addition, these technologies will also speed up and securitise the process of buying, selling and trading stock with great transparencies and auditability.

There are many possibilities for cryptocurrencies, contract-based activities like wills, accounting, insurance, travel and car-tracking and car-leasing, trucking and shipment-tracking, the streamlining of parts inventories, direct provider-to-consumer interactions in hospitality, industrial IoT, 3-D printing, improvements to project management in architecture and real estate, as well as healthcare management and more effective research and clinical trials. This is not to mention the ways blockchain tech can provide paperless records and better data management especially in gun tracking and law enforcement. The uses expand in relation to national mail system, public transportation, waste management, retails systems and e-commerce more broadly, agriculture, logging and mining. In academia blockchain offers a streamlining of verification processes and the prospect of the development of new platforms outside the big publishers to enable greater collaboration. Libraries, in particular, can expand services through metadata archives. It seems likely that costs of video streaming can be heavily reduced through decentralized coding, storage and distribution. There are innumerable uses in gaming, gambling, art, photography as well as cloud computing and storage, internet identity, advertising and human resources.[6]

The digital economy partnership agreement (DEPA)

The DEPA Partnership Agreement is described as 'A new partnership between New Zealand, Chile and Singapore will help New Zealand exporters and SMEs take advantage of opportunities from digital trade.'[7] It 'entered into force for New Zealand and Singapore on 7 January 2021' and was signed in an 'online virtual signing ceremony' on 12 June 2020. The Report of the Economic Development, Science and Innovation Committee (2020) emphasized that digital trade had opportunities for small and medium-sized businesses where the new technologies help them 'to overcome the challenges of scale and distance to enter global markets previously only accessible to larger businesses.'[8] New Zealand is the 'depository' for the DEPA and

> will receive any future applications for further state accessions. This will place New Zealand at the heart of the DEPA's future as a living agreement, and will allow New Zealand to set the agenda for accession.[9]

Already China, South Korea and Canada[10] has signalled their interest in joining. The report sets out the reasons for NZ to join including business and trade facilitation, emerging trends and technologies and innovation and the digital economy. The Executive Summary of the National Interest Analysis states:

> Conclusion of the Digital Economy Partnership Agreement (DEPA or the Agreement) comes at a time of considerable disruption to international trade and supply chains due to COVID-19. New Zealand's immediate goal has been to bring the pandemic under control within our borders and to mitigate as far as possible its negative impacts on our people, health systems and the economy. Work is also under way on New Zealand's trade recovery, to help ensure that New Zealand is in the best possible position to emerge from the COVID-19 pandemic as quickly as possible….
>
> Digital trade (digitally-enabled transactions of goods and services, whether digitally or physically delivered) has grown rapidly, racing ahead of existing global trade rules and norms. In the context of the COVID-19 pandemic, this takes on even greater significance as these rules will be more important than ever in assisting with the global economic recovery. The DEPA is a first step towards establishing those rules and best practice for the digital era. It contains a series of modules covering topics relevant to the digital economy and digital trade.

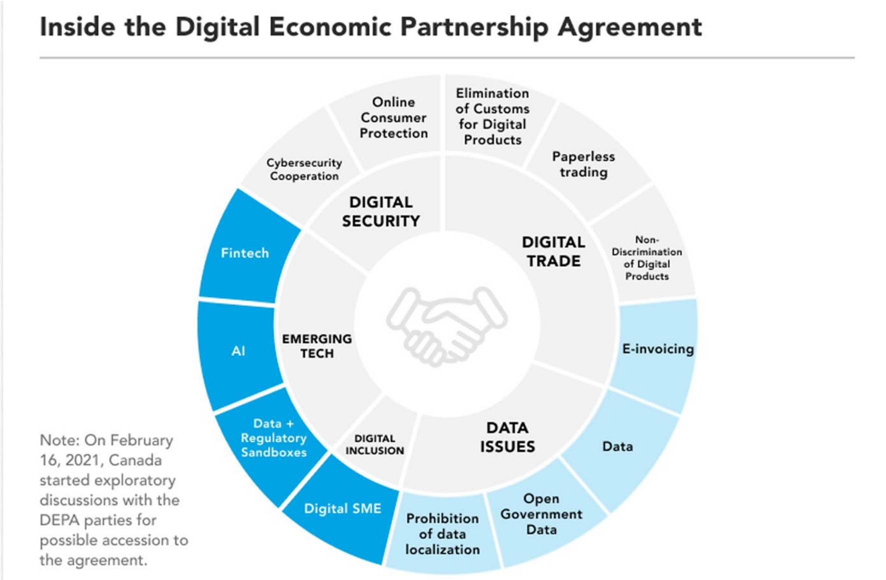
The DEPA agreement indicates that NZ's objectives are to: '*Co-create and shape global norms for digital trade; Create a model digital economy agreement that can act as a pathfinder; and Build confidence on new economy issues*' (italics in original).

Taheri, Adams, and Stern (2021) in their article 'DEPA: The World's First Digital-Only Trade Agreement' indicate the stakes:

> in 2020, retail e-commerce sales in Canada increased by 70.5 per cent compared to 2019…
>
> Today, the digital economy is estimated to account for as much as 15.5 per cent of global GDP, or $C14.35 trillion.

> In Canada, digital economic activities have grown exponentially over the past decade to reach upwards of an estimated 5.5 per cent of national GDP.[11]

They provides a useful graphic that illustrates the main 'modules' (chapters) of the agreement



As recently reported 'At the 2021 G20 Summit in Rome, President Xi Jinping announced that China would seek to join the newly created Digital Economy Partnership Agreement (DEPA). A day later, on November 1, China's Ministry of Commerce (MOFCOM) formally applied for China to join the agreement.'[12] The Ministry of Commerce of the People's Republic of China reported:

> Applying to join DEPA is in line with China's direction of further deepening domestic reforms and expanding high-level opening to the outside world, and will help China strengthen cooperation with all members in the field of digital economy under the new development pattern, and promote innovation and sustainable development. In the next step, China will carry out follow-up work with all members in accordance with relevant DEPA procedures.[13]

The interest of China, South Korea and Canada in joining DEPA in conjunction with membership of The Regional Comprehensive Economic Partnership (RCEP) indicates the the importance of digital trade and how it has become a major force in the Asia-Pacific regional economy with great potential for scientific research collaboration and international education. It is clear that digital trade is the future of trade and investment among the most dynamic in the world economy able to harness blockchain technologies, artificial intelligence and the Internet of things to guide the expansion of e-commerce and cross-border payments. The World Economic Forum has developed a list of digital trade priorities based on accelerating digital trade preparedness, developing and promoting the interoperability of 'global data flows, including through trade frameworks and regulatory cooperation', building a new framework for cross-border digital services, and mapping new trade technologies 'including cloud services, DLT and 3D printing'.[14]

ASEAN 'is the fastest growing Internet market in the world. With 125,000 new users coming onto the Internet every day, the ASEAN digital economy is projected to grow significantly, adding an estimated $1 trillion to regional GDP over the next ten years.'[15] Digital ASEAN is a good example of both the massive potential and the challenges with four workstreams: 1. Pan-ASEAN Data Policy: shaping a common regional data policy; 2. ASEAN Digital Skills: building a shared commitment to train digital skills for the ASEAN workforce; 3. ASEAN e-Payments:

building a common ASEAN e-payments framework; 4. ASEAN Cybersecurity: nurturing cooperation and capacity building in ASEAN cybersecurity.[16] 'ASEAN Digital Skills Vision 2020: Training the ASEAN Workforce for the Fourth Industrial Revolution' launched in 2018 provides a pledge framework to train 20 million ASEAM SME workers and 40 million citizens in digital literacy.[17]

A recent NZIER study 'found that if digital trade improved along New Zealand's supply lines, both it and its Asia-Pacific Economic Corporation trading partners could gain up to $18 billion over the next decade'.[18] DEPA is just the beginning and NZ is taking a leading role. In this framework more attention should be paid to scientific research and development, academic publishing, and the education economy as a means of promoting innovation in digital trade.

Notes

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